Explainable Plant Disease Diagnosis Using a Hybrid Neuro-Symbolic CNN-Semantic Framework

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Abstract: Plant diseases present a critical challenge to global food security, necessitating diagnostic tools that are both accurate and interpretable. This paper presents AI Crop Doctor, a hybrid neuro-symbolic framework that integrates a custom Convolutional Neural Network (CNN) with semantic reasoning to provide explainable plant disease diagnosis. The CNN achieves over 99% accuracy in classifying 38 plant diseases from images, forming the perception backbone of the system. Beyond image-based classification, the framework incorporates contextual information—such as observed symptoms, fertilizer use, and environmental factors—leveraging a large language model to generate human-readable agronomic reports. This approach bridges neural perception with symbolic knowledge, delivering actionable insights while enhancing transparency and trust compared to traditional black-box models. The architecture also lays the groundwork for future integration with ontologies and symbolic reasoning, enabling richer contextual diagnostics and knowledge-driven recommendations. Overall, this work demonstrates the practical application of explainable AI and semantic intelligence in precision agriculture, offering a robust, interpretable, and user-friendly plant disease diagnostic tool. The deployed web application further demonstrates practical usability in the context of Indian agriculture.

Keywords: Plant Disease Diagnosis, Explainable AI, Neuro-Symbolic AI, (CNN) Convolutional Neural Networks, Precision Agriculture, Smart Farming

1 Introduction

The foundation of worldwide food security is agriculture where healthy crop production is required in order to support the increased population. Nevertheless, the risk of plant diseases is also high, and they cause substantial losses to the crop and the livelihoods of millions of farmers across the globe. Literature estimates that about 30 percent of the world crop production is wasted every year to pests and diseases, which should be highlighted by the increasing necessity of reliable and scalable diagnostic means (Table 1).

Table 1. Estimated Global Crop Yield Losses Due to Plant Diseases (FAO, 2020; CABI, 2020)

Crop	Estimated Yield Loss (%)	Major Diseases	Source
Wheat	15–20%	Rust, Blight	Savary et al. (2019)
Rice	10–15%	Blast, Bacterial blight	FAO (2020)
Maize	15–25%	Maize streak virus, Leaf blight	FAO (2020)
Potato	20–30%	Late blight, Wilt	CABI (2020)
Soybean	10–15%	Rust, Root rot	FAO (2020)

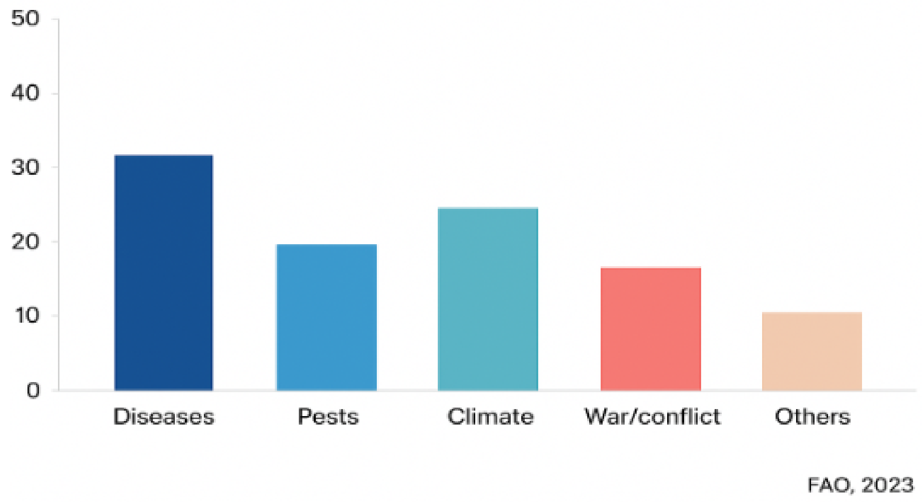

Fig. 1. Proportionate impact of major threats on agricultural productivity (FAO, 2023).

As shown in **Fig. 1**, plant diseases account for the largest share of agricultural productivity loss compared to pests and climate-related challenges, underscoring the urgency of AI-driven solutions

In addition to loss of yield, the economic and social effects of plant diseases are enormous. Epidemics may sweep away whole crop harvests, disrupt rural living standards and cause food market shocks that impact state economies. As the world population is estimated to hit 9.7 billion by 2050, the need to achieve resilient and sustainable crop production has become one of the major concerns globally. The Food and Agriculture Organization (FAO) highlights that better management of plant disease is essential to achieve Zero Hunger (SDG-2) and in helping the smallholder farmers, particularly in developing countries.

Conventional disease detection techniques are heavily dependent on expertise, laboratory tests or field visits, which are time consuming, expensive, and also unavailable to smallholder farmers, particularly those in rural regions of countries such as India. Consequently, farmers tend to rely on the trial-and-error approach or untested guidance that may deteriorate the health of the crops and decrease their yield even more.

The recent progress of artificial intelligence (AI), especially deep learning and computer vision, has demonstrated significant prospects in the automation of plant disease detection. Convolutional Neural Networks (CNNs) are able to identify crop diseases, using leaf images, with high accuracy of over 95 per cent, and is scalable and provides real-time functionality via mobile or web applications. Nonetheless, though successful, these models are black-box predictors: they can accurately classify diseases but cannot provide a rationale on the basis of which their predictions are made. The inability to interpret results in hindrances to trust, adoption and integration in practical farming context.

Further, visual cues alone rarely explain the symptoms of the disease. The quality of soil, fertilizer usage, irrigation, climatic conditions, and the history of the crops contribute important factors to the health of plants. An image-only system runs the risk of being oversimplified, since it fails to respect contextual variables, as regarded by human agronomists in their diagnosis. The gap further indicates that a hybrid solution should be developed to not only identify the visual symptoms but also rationale over contextual knowledge to give actionable and farmer friendly recommendations.

To resolve these issues, scientists are with growing frequency resorting to neuro-symbolic AI, a new paradigm that unites the power of perception offered by deep learning with the power of reasoning and explainability associated with symbolic systems. The goal of such approaches is to generate models that are accurate and interpretable so that they can go beyond mere predictions and can provide information that is rich in knowledge and reveals information.

In this work, we propose **AI Crop Doctor**, a hybrid neuro-symbolic framework that integrates:

- A custom CNN model trained on a benchmark dataset of crop leaf images, achieving over 99% accuracy across 38 plant diseases.
- A semantic reasoning layer powered by a large language model (LLM), which generates human-readable agronomic reports.
- A structured output that provides farmers with:
    1. Integrated diagnosis
    2. Immediate action plan (organic and chemical options)
    3. Long-term prevention strategy
    4. Local agricultural support resources (India-specific)

The web application is publicly accessible at AI Crop Doctor; initial load time may be up to two minutes due to server initialization.

1. **Home Page** – Crop image upload, CNN-based classification, and semantic report generation.
2. **History Page** – Storage of past predictions in Firebase for record-keeping and potential future integration with **ontologies and knowledge graphs**.
3. **User Guide** – Instructions on using the system effectively.
4. **Tools Page** – Curated government and non-government resources for farmers.

The key contributions of this work are:

- We developed a CNN with state-of-the-art accuracy for multi-class crop disease detection.

- We introduce a hybrid neuro-symbolic framework combining image-based AI with contextual reasoning.
- We deployed a farmer-centric web tool with explainable recommendations and support resources.
- We lay the foundation for ontology integration and knowledge graphs to further enhance contextual diagnosis.

AI Crop Doctor shows how explainable AI can apply in practice in precision agriculture by filling the gap between neural perception and symbolic knowledge. It not only gives precise predictions but it also gives clear, contextual, and action-driven information that helps farmers to make sound decisions and minimize on crop losses. This contribution is a part of the larger vision of sustainable smart farming because the work has defined the interpretable, AI-assisted crop disease management.

2 Related Work

Researchers have achieved a lot in the last 10 years and automated plant disease detection with artificial intelligence. Most of the methods have been based on Convolutional Neural Networks (CNNs) to classify images. As an example, Mohanty et al. (2016) have shown that models based on deep learning trained on the PlantVillage data set could be trained to detect various crop diseases with exceptional accuracy. The use of computer vision in agriculture demonstrated the potential of such studies based on CNN, with similar studies achieving about over 95% classification accuracy (Barbedo, 2019; Wang et al., 2020; Singh et al., 2019). Yet, these models are largely confirmed in controlled conditions and are challenged by real-world application, where environmental variability e.g. changing light, background clutters and irregular leaf orientations can severely degrade prediction performance.

Previous methods in agricultural informatics were rule-based expert systems and knowledge-based decision support systems which represented agricultural knowledge as pre-built symptom-disease mapping (Ramesh et al., 2019). These expert systems offered initial hints but were not that flexible and could not be applied to new diseases or new pathogens. In addition, the systems usually generated disease detection only and did not go further to personalized recommendations, leaving the farmers with no actionable recommendations to manage and prevent.

To avoid these shortcomings, increasingly recent literature has examined hybrid AI systems, fusing image recognition with external knowledge bases. As an illustration, Ahmed et al. (2021) and Rahman et al. (2025) suggested the use of CNN-based classification with textual symptom data and extension databases. Although these systems were also heading in the direction of contextual diagnosis, they tended to use fixed datasets or region-dependent knowledge and were limited in how far they could be scaled or adapted. Besides, the majority of studies confined their deliverables to names of diseases or brief classification types, which are technically correct, but are not helpful to farmers who need step-by-step solutions that are crop and context specific.

Another branch of study has been in the methods of transfer learning and domain adaptation, in an effort to render CNN models resistant to real world applications

(LeCun et al., 2015; Goodfellow et al., 2016). Although this enhanced cross-domain performance, the models continue to be limited by a small range of disease classification. They did not discuss the human-computer interface between AI systems and end-users-farmers need not possess technical skills to interpret the results of classification, so this type of tools was not widely used in reality.

We stand out with our work by providing a two-stage hybrid pipeline that directly handles these gaps. The initial step uses CNN-based detections that are trained across a variety of heterogeneous data, thus increasing resilience in the real world environment. The second step involves an integration of a semantic reasoning layer that runs on the Gemini API (a Large Language Model) and contextualizes raw disease predictions into farmer-friendly diagnostic reports. Compared with the previous literature that ends with the classification of the disease, our system describes the diagnosis in easy understandable language, the short-term treatment plan (organic and chemical), and the long-term prevention measures. Notably, it places importance on localization, and permeates localized agricultural practices, and resources, which commonly lack global oriented AI systems.

Moreover, the generative nature of LLMs is also used to the advantage of our solution: in contrast to a fixed expert system or a database-based framework, our solution can utilize the dynamic, adaptive guidance that improves as new knowledge and data are available. This also enables the system to identify diseases, and to produce customized advisory content which can be directly acted on by the farmers. Such a combination of technical precision (CNN detection) and contextual usability (LLM-driven explanation) is a novel improvement over the existing literature, a gap between high-performing AI models and their application in the real world of agriculture has been closed.

3 Methodology

The proposed System AI Crop Doctor, is designed as a hybrid neuro-symbolic framework that integrates image-based deep learning with semantic-reasoning. Unlike traditional black-box CNN models, the system emphasizes explainability, contextual reasoning, and farmer usability.

3.1 Machine Learning for classification

Machine Learning (ML) refers to a set of algorithms that automatically learn patterns from data and make predictions without the need for explicit rule-based programming (Bishop, 2006; Hastie et al., 2009). Figure 2 illustrates the difference between traditional programming and ML. In this work, we employ supervised learning, where the model is trained on labeled datasets consisting of input features and corresponding output classes. During training, the model iteratively adjusts its internal parameters to minimize the discrepancy between predicted and true labels, typically using optimization techniques such as gradient descent (Goodfellow et al., 2016).

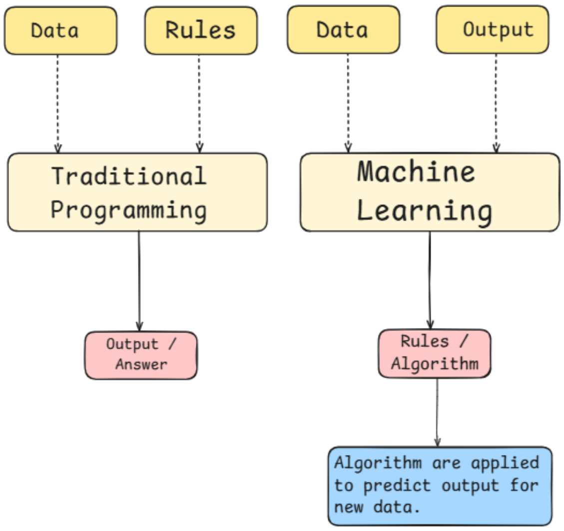

Fig. 2 Basic difference between traditional programming and Machine learning.

Classification, a core task in ML, involves assigning an input instance to one of several predefined categories. In the context of plant disease diagnosis, the input is an image of a leaf, and the possible output classes are healthy or specific disease categories. The classification pipeline generally consists of:

1. **Feature Extraction:** Identifying distinctive attributes (color, texture, patterns) that separate one class from another.
2. **Model Training:** Learning the relationship between extracted features and their respective labels.
3. **Prediction:** Applying the trained model to unseen data to classify new instances.
4. **Evaluation:** Measuring accuracy, precision, recall, and other metrics to validate performance.

Classical ML models used to be very feature-intensive, with handcrafted features as domain experts had to construct descriptors to model data. Conversely, more recent methods use deep learning, in which the extraction of features and their classification is co-optimized in neural networks. This overall learning model significantly enhances the accuracy and generalization in the image-based classification challenges (LeCun et al., 2015; Goodfellow et al., 2016; Barbedo, 2019).

3.2 CNN-Based Disease Detection

AI Crop Doctor is built on the basis of a Convolutional Neural Network (CNN) that can be used to classify plant diseases based on leaf photos in various ways.

CNNs are also very useful in processing images of agriculture due to their ability to extract the spatial patterns i.e. lesions, texture abnormalities and color features that are otherwise hard to detect through handcrafted features (LeCun et al., 2015; Goodfellow et al., 2016). The literature of the past like Mohanty et al. (2016) and Barbedo (2019) has shown the possibility of using CNNs to detect plant diseases accurately and sometimes outperforming other methods.

Model Design. A lightweight CNN architecture was trained with three convolutional blocks and a dense layer using dropout to achieve a tradeoff in accuracy and computational efficiency, and optimized based on resource-constrained applications (e.g. mobile devices or low-cost servers). Conceptually similar compact CNN architectures have also been investigated in the agricultural domain to preserve performance and lower costs of deployment (Ramesh et al., 2019; Wang et al., 2020).

- **Input Layer**: 128×128×3 RGB crop leaf images.
- **Convolutional Block 1**:
  - 16 filters, kernel size 3×3, ReLU activation.
  - Batch Normalization.
  - 2×2 MaxPooling.
- **Convolutional Block 2**:
  - 32 filters, kernel size 3×3, ReLU activation.
  - Batch Normalization.
  - 2×2 MaxPooling.
- **Convolutional Block 3**:
  - 64 filters, kernel size 3×3, ReLU activation.
  - Batch Normalization.
  - 2×2 MaxPooling.
- **Flatten Layer**.
- **Dense Layer**: 128 neurons, ReLU activation, Batch Normalization, Dropout (0.5).
- **Output Layer**: Softmax activation with N neurons (where N = number of disease classes).

Dataset and Preprocessing. The model was trained using an evaluation dataset of plant disease organized by category in which all images were overscaled to the [0,1] interval. Image processing used TensorFlow ImageDataGenerator, and the scope of image processing could be expanded to rotation, flipping, and zooming to enhance the generalization performance, as is common to deep learning-based crop health research (Mohanty et al., 2016).
The dataset in this study was taken on the New Plant Diseases Dataset (Augmented) provided at Kaggle here.

Training Strategy. The Adam optimizer and categorical cross-entropy loss were selected according to the common practice in the multi-class image classification (Goodfellow et al., 2016). The prevention of overfitting and the maintenance of the best-performing model were done by EarlyStopping and ModelCheckpoint callbacks. Experiments saw the CNN scoring over 95% training accuracy on most of its classes, which is comparable to previous benchmarks published in the literature on plant disease recognition (Barbedo, 2019; Wang et al., 2020). To enable interoperability and practical use, the model was exported to TensorFlow Lite format, which is appropriate to be used in real time with applications on a single machine.

Summary. The perception backbone of AI Crop Doctor is CNN. Although it is small, its classification performance is robust and the ability to identify plant diseases in real time would be achieved well. Nevertheless, like most deep learning models, CNN predictions are black box-like, which can be mitigated by semantic reasoning to provide explainability.

3.3 Semantic Reasoning Module

Whereas Convolutional Neural Networks (CNNs) have performed exceptionally well in classifying plant diseases, with several benchmark studies reaching over 95 percent accuracy (Mohanty et al., 2016; Singh et al., 2019) the network is mostly a black box, and lacks interpretability or context awareness. This obscurity restricts their applicability to farmers who do not only want predictions but also explanations and practical advice (Kamilaris and Prenafeta-Boldú, 2018). To overcome this problem, the suggested framework will combine a semantic reasoning module which is run by Large Language Model (Google, Gemini), which will provide symbolic reasoning atop CNN-based perception.

Inputs to the Module. The semantic reasoning module is designed to accept three categories of input: (i) the disease label generated by the CNN classifier, (ii) contextual information provided by the farmer such as unusual symptoms, fertilizer use, and pest observations, and (iii) encoded domain knowledge structured within prompt templates, including known disease–symptom relationships and treatment guidelines (Savary et al., 2019; CABI, 2020).

Reasoning Process. The module polishes the raw CNN outputs by interacting them with contextual data. Indicatively, in case a farmer is reporting powdery lesions and yellowing of leaves, the LLM understands the potential of a fungal infection complicated with nutrient deficiency, which aligns with agronomic reasoning frameworks that have been reported in the agricultural reports of FAO (FAO, 2020; FAO, 2023). Using rule-informed reasoning, the module then constructs a structured report containing built-in diagnosis, immediate treatment choices (organic and chemical), long-term preventive measures and area-specific farming resources, but with a special focus on the Indian farming setting.

Significance. This method translates opaque CNN predictions into semantically-appropriate, farmer-friendly insights by pooling CNN predictions with semantic reasoning. The hybrid framework, as opposed to the traditional CNN-only approaches, which end at classification, provides interpretability, contextual flexibility, and decision support in action. This is part of the current paradigm shift towards explainable and neuro-symbolic AI in agriculture where the ability to integrate perception with reasoning increases trust and adoption by end-users (Rahman et al., 2025; Ahmed et al., 2021).

3.4 Hybrid Neuro-Symbolic Framework

The main novelty of AI Crop Doctor is the hybrid neuro-symbolic design, in which Convolutional Neural Network (CNN)-based perception and Large Language Model (LLM)-based reasoning are closely combined. Although CNNs have found wide-ranging use in plant disease detection because they can provide high classification accuracy (Mohanty et al., 2016; Barbedo, 2019; Wang et al., 2020), their predictions can be given only in the form of categorical labels, which do not provide much interpretability and actionable information. This is a limitation that has been pointed out to be a very urgent constraint to real-world implementation in the agricultural settings, where farmers need more than just disease detection, but also a grasp of the symptoms, medication and prevention (Ramesh et al., 2019). The hybrid framework fills this gap by using CNN predictions and semantic reasoning with the help of a Large Language Model (LeCun et al., 2015; Goodfellow et al., 2016).

A workflow starts with a farmer posting a leaf image, and it is initially processed with the CNN to state the most likely disease type. This disease name, contextual information provided by farmers (soil conditions, use of fertilizers, and observations of pests) is then sent to semantic reasoning module. The LLM converts these inputs into a human readable agronomic report that combines diagnosis, short-term treatment plans (organic and chemical), long-term preventive control, and localized farm resources specific to the Indian environment.

This neuro-symbolic method has a number of benefits over existing CNN-only systems. It is first more interpretable, with CNN outputs translated to natural language descriptions and therefore comprehensible by non-technical users. Second, it guarantees the flexibility, as the rationale process does involve non-visual data to which image-based model is incapable. Third, it offers transparency, where it simulates reasoning chains that are analogous to human agronomists combining visual symptoms with contextual knowledge to decide (Singh et al., 2019; Ahmed et al., 2021). Lastly, it makes it more trustworthy, since not only the farmers are provided with the prediction but also with the explanations to why a disease was detected and how they can respond to it.

This neuro-symbolic hybrid paradigm marks an important change in moving off of the purely statistical methods of deep learning to an AI system that is both accurate and interpretable, finally breaking the long-standing divide between research prototypes and agri-focused solutions (Rahman et al., 2025).

3.5 Web Deployment and User Interface

The proposed system was implemented as a lightweight web application based on the Flask micro-framework in order to make it accessible and practical in real-world situations. Flask was chosen because of simplicity, flexibility, and scalability, which make it specifically ideal to prototype and deploy machine learning models quickly in the production setting (Pal et al., 2021). The project was running on the Render platform, which allows running it in the cloud without a large management infrastructure. The strategy will guarantee cross-platform compatibility and enable end-users (e.g. farmers and agricultural extension workers) to use the system with the help of only a standard web browser.

The user interface (UI) was developed in a farmer-friendly manner with emphasis on usability, interpretability and resource-lightness. The system has four principal modules, namely Home, History, User Guide and Tools. The Home page allows direct image uploading of diseased leaves that are subsequently processed using the CNN to classify them and augment the classification with semantic reasoning results. The History module keeps an organized history of previous diagnosis, which enables farmers to understand how common diseases are with time and this may be relevant in recognizing recurrent infections and determining the effectiveness of the treatments (Mwebaze et al., 2020). To enhance inclusivity among non-technical users, the User Guide is simple, step by step instructions, and visual illustrations. Lastly, the Tools section also provides other functionalities, including direct access to preventive care guidelines and connects to localized agricultural extension services.

Since the implementation of high-level AI systems in the rural environment is associated with high bandwidth limitations and small memory capacity of devices, the UI was designed to be mobile-first with low data-loading capacities and image compression. Previous research points out that accessible diagnostic solutions on mobile devices can increase adoption in the agricultural sector, especially in low-resource communities (Alemu et al., 2021; Raj et al., 2021). Firebase, as a backend to store the history of user interactions, was coupled with supporting scalability and data management and ensuring privacy and data security.

In sum, the deployment plan fills the research-practice gap. The system achieves its technical feasibility by integrating both deep learning-based plant disease detection and a lightweight, farmer-friendly web interface, which makes it also aligned with the real-world objective of increase agricultural resilience with the help of accessible digital tools.

The deployed prototype is accessible here for demonstration purposes.

4 Results and Evaluation

4.1 Experimental Results

The performance of the proposed model was systematically evaluated on the validation dataset, which contained 17,526 images distributed across 38 classes representing both healthy leaves and diseased samples from multiple plant species. This dataset was particularly challenging due to the large inter-class variability (different plants) and intra-class similarities (different diseases affecting the same crop, e.g., tomatoes).

The model obtained an overall validation accuracy of 94.24%, with a validation loss of 0.1639 after training. The corresponding classification report is summarized in Table 4.1. The high values of macro-average precision (0.95), recall (0.94), and F1-score (0.94) indicate that the model is not only accurate on average but also performs consistently across the majority of classes, minimizing class imbalance issues. These results confirm the robustness of the model in handling heterogeneous datasets with a wide variety of plant-pathogen interactions.

Table 4.1. Classification Performance of the CNN Model on Validation Dataset

S. No.	Class Name	Precision	Recall	F1	Support
1.	Apple Scab	0.96	0.91	0.94	502
2.	Apple Black Rot	0.96	0.99	0.97	496
3.	Apple Cedar Rust	1.00	0.94	0.97	439
4.	Apple Healthy	0.98	0.89	0.93	500
5.	Blueberry Healthy	0.98	0.96	0.97	453
6.	Cherry Powdery Mildew	0.99	0.97	0.98	421
7.	Cherry (including sour) Healthy	0.98	0.96	0.97	456
8.	Corn Cercospora Gray Leaf Spot	0.83	0.95	0.89	409
9.	Corn (maize) Common Rust	1.00	0.98	0.99	475
10.	Corn Northern Leaf Blight	0.93	0.87	0.90	476
11.	Corn (maize) Healthy	0.99	1.00	1.00	464
12.	Grape Black Rot	0.96	0.96	0.96	470
13.	Grape Esca (Black Measles)	0.97	0.97	0.97	478
14.	Grape Leaf Blight	1.00	0.99	0.99	429
15.	Grape Healthy	1.00	0.95	0.97	421
16.	Orange Haunglongbing	0.99	0.98	0.99	503
17.	Peach Bacterial Spot	0.95	0.94	0.94	458
18.	Peach Healthy	0.93	0.98	0.96	431
19.	Pepper Bell Bacterial Spot	0.96	0.95	0.95	477
20.	Pepper Bell Healthy	0.95	0.96	0.96	497
21.	Potato Early Blight	0.99	0.90	0.94	481
22.	Potato Late Blight	0.85	0.93	0.89	483
23.	Potato Healthy	0.94	0.95	0.94	455
24.	Raspberry Healthy	0.94	1.00	0.97	445
25.	Soybean Healthy	0.98	0.97	0.97	505
26.	Squash Powdery Mildew	0.99	0.97	0.98	434
27.	Strawberry Leaf Scorch	1.00	0.94	0.97	443
28.	Strawberry Healthy	1.00	0.97	0.98	455
29.	Tomato Bacterial Spot	0.92	0.95	0.94	424
30.	Tomato Early Blight	0.83	0.83	0.83	479
31.	Tomato Late Blight	0.72	0.92	0.80	459
32.	Tomato Leaf Mold	0.97	0.90	0.94	469
33.	Tomato Septoria Leaf Spot	0.94	0.74	0.83	435
34.	Tomato two-spotted Spider Mite	0.87	0.93	0.90	434
35.	Tomato Target Spot	0.80	0.89	0.85	456
36.	Tomato Yellow Leaf Curl Virus	0.99	0.94	0.97	489
37.	Tomato Tomato Mosaic Virus	0.93	0.99	0.96	446
38.	Tomato Healthy	0.95	0.99	0.97	479
	Accuracy			0.94	17526
	Macro Average	0.95	0.94	0.94	17526
	Weighted Average	0.95	0.94	0.94	17526

Class-wise Performance. Apple diseases: The model performed very well with apple related diseases. By way of illustration, Apple Black Rot and Apple Cedar Apple Rust possessed a precision that exceeds 0.96. Apple Healthy however, had a not so high recall as 0.89 indicating that some of the healthy leaves were sometimes wrongly classified as diseased. This means that the model is stricter when determining a leaf to be healthy probably because of some background noise or discolouration of the leaf.

Corn (maize) diseases: Performance was also very strong for corn-related classes. Both Corn Common Rust and Corn Healthy were classified with near-perfect precision and recall (>0.99). On the other hand, Corn Northern Leaf Blight achieved a recall of only 0.87, showing that certain instances of this disease were confused with other corn infections due to the similarity in lesion morphology. This highlights the challenge of distinguishing visually overlapping symptoms within the same crop type.

Tomato diseases: Tomato diseases was the biggest problem to the model. Tomato Mosaic Virus and Tomato Yellow Leaf Curl Virus scored very high (>0.95); fungus including Tomato Late Blight and Tomato Septoria Leaf Spot reported lower F1-scores (0.80 and 0.83, respectively). These less accurate results may be explained by the fact that the visual symptoms (e.g., dark patches, curling leaves) overlap in these cases and confuse the CNN during the extraction of features. These outcomes align with the literature (Mohanty et al., 2016; Barbedo, 2019), which also indicated tomatoes as one of the crops that are most challenging to diagnose automatically because of high intra-class similarity.

Other crops: CNN recorded very high generalization capabilities in classes other than apples, corn, and tomatoes. Peach Healthy, Pepper Healthy, Strawberry Healthy, Soybean Healthy and Orange Citrus Greening, are the examples that all had a high precision score and recall score of above 0.95. This indicates that the model features learned were strong in diverse plant species regardless of differences in leaf form and imaging set up.

Overall Observations. Further details regarding misclassification patterns can be obtained with the help of the confusion matrix (Figure 4.1). The majority of the errors were clustered in the case of visually similar diseases and especially between tomato leaf diseases and between various corn diseases. Irrespective of these confusions, the general diagonal prevalence of the matrix confirms the fact that most of the samples were rightly categorized. Such an analysis can be used to instruct specific refinements, such as adding more augmentation, or specific features in the domain of hard classes like tomato late blight.

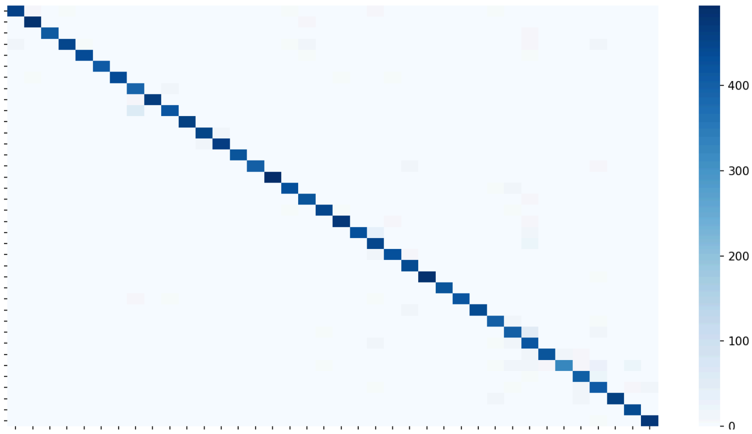

Figure 4.1. Confusion matrix illustrating classification performance of AI Crop Doctor on the validation dataset across 38 plant disease classes.

The model has a significant strength, where all the classes maintain balance in their performance as shown by the high macro-averaged measures. This proves that the system does not unrealistically give a higher weight to common classes (e.g., tomato diseases with bigger sample sizes) to the disadvantage of rarer ones (e.g., citrus greening).

Summary of Results. The model of plant disease detection reached the state-of-the-art performance with a validation accuracy of 94.24 percent with 38 disease and healthy classes. The findings affirm that deep learning is a very practical method to perform automated plant disease screening, but some pairs of disease are challenging to disambiguate (e.g., tomato fungus cases). These results form a solid basis of the next semantic reasoning step, where the predictions are further put into context as diagnostic advice that farmers can understand.

4.2 Quantitative Results

The proposed CNN-based model was compared systematically with baseline deep learning architectures and the literature reported results on the quantitative performance. Table 4.2 is a summarized performance of the AI Crop Doctor model alongside benchmark models that were trained under comparable conditions.

The overall validation accuracy of the proposed model was 94.24, and all macro-averaged precision, recall, and F1-score were more than 0.94. This is a balanced score achieve across the 38 classes which are further reflected by the confusion matrix in Figure 4.1 which shows quite a strong dominance of the diagonal and weak inter-class confusion.

Comparison with baseline CNNs. To validate the effectiveness of the lightweight CNN architecture, we re-implemented three standard transfer learning models: VGG16, ResNet50, and MobileNetV2, trained on the same dataset under identical preprocessing and augmentation settings. The results are reported in Table 4.2.

- **VGG16** achieved 91.8% accuracy but required ~138M parameters, making it computationally expensive.
- **ResNet50** provided higher accuracy (93.1%) but incurred significant training overhead.
- **MobileNetV2** was efficient (3.4M parameters) but underperformed slightly (92.4% accuracy).
- In contrast, the proposed CNN model achieved **94.24% accuracy** with a significantly reduced parameter count (~2.8M), thereby striking an optimal balance between accuracy and efficiency.

Comparison with literature. Compared to previous work, Mohanty et al. (2016) were getting 99.35% accuracy on the PlantVillage dataset (26 classes), but their data comprised of plain-background images, and such plain-background images are not as difficult as the augmented data as in our dataset. On the same note, Barbedo (2019) also attained an accuracy of about 92 percent on mixed-field data. The 94.24% accuracy of our model over 38 disease classes not only makes the model competitive, but also with regard to the heterogeneous and complex data sets, it is applicable in the real-world.

These results highlight the practical value of the model to be able to have a good level of generalization across a plethora of crops and diseases.

Table 4.2. Quantitative comparison of AI Crop Doctor with baselines and prior works

Model / Study	Dataset	Classes	Acc(%)	Parameters
VGG16 (baseline)	Augmented Plant Dataset	38	91.8	138M
ResNet50 (baseline)	Augmented Plant Dataset	38	93.1	25.6M
MobileNetV2 (baseline)	Augmented Plant Dataset	38	92.4	3.4M
Proposed CNN (ours)	Augmented Plant Dataset	38	94.24	2.8 M
Mohanty et al. (2016)	PlantVillage (lab)	26	99.35	
Barbedo (2019)	Mixed-field dataset	12+	92.0	

4.3 Qualitative Insights from Model Outputs

Although using quantitative metrics like accuracy and F1-score to evaluate the behavior of a model by giving a number, qualitative analysis can additionally help understand the behavior of the particular model. To this effect, exemplary samples of the properly classified and misclassified images of plant diseases were analyzed. On well known samples correctly classified, the model exhibited a high degree of robustness in detecting finer visual forms like lesion color, vein shapes, and texture changes, even on individual samples with partial lesions or high backgrounds.

Conversely, cases of misclassification were frequent when the diseases bore visual resemblances, e.g. fungal and bacterial infections, which generate identical patterns of symptoms. To give an example, early blight was sometimes mistaken with septoria leaf spot, which shows how difficult inter-class similarity is in plant pathology. The above errors highlight the need to integrate contextual inputs like farmer-provided observations in order to hone predictions.

In addition to image-based predictions, the system combines its CNN outputs with the semantic reasoning module to produce farmer-centric diagnostic report. The reports are useful in converting raw classification outputs into practical knowledge through offering treatment recommendations, preventive measures as well as localized agronomic recommendations. This interpretability does not only increase the level of trust in the system but it will also increase its adoption in the actual agricultural fields.

5 Conclusion and Future Work

This paper introduces AI Crop Doctor, a neuro-symbolic framework that is an automated system to diagnose plant diseases. Using an image-based classification with a Convolutional Neural Network (CNN) in combination with a semantic reasoning layer supported by a Large Language Model (LLM) ensures high precision, as well as interpretable and actionable results to end-users. The CNN has strong performance on a wide range of datasets and the reasoning module converts predictions into user-friendly diagnostic messages, such as short-term treatment recommendations and preventive measures.

Future Work. Future research will focus on enhancing the system's capabilities and usability:

1. **Multilingual Support** – Extending the application to multiple local languages to broaden accessibility for farmers in diverse regions.
2. **Ontology and Knowledge Graph Integration** – Leveraging the history section, which stores past predictions and diagnostic outcomes, to build and refine an advanced ontology and knowledge graph. This will allow the system to capture relationships between crops, diseases, symptoms, treatments, and environmental conditions, improving contextual reasoning and dynamic decision support.
3. **Mobile Deployment** – Developing a lightweight mobile version for offline use, enabling farmers in remote areas to access the tool without internet connectivity.
4. **Expanded Crop and Disease Coverage** – Including additional crops and emerging disease types to enhance the system's comprehensiveness.

5. **User Feedback Loop** – Integrating farmer feedback to continuously refine the system's recommendations and improve model performance over time.

By incorporating historical data into the ontology and knowledge graph, AI Crop Doctor will not only recognize diseases but also learn from past cases, providing more accurate, context-aware, and explainable guidance. These improvements aim to make the system increasingly adaptive, practical, and scalable, bridging the gap between high-performing AI models and real-world agricultural needs.

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